PERURI DIGITAL SUMMIT 2026
HACKATHON

IGOR: Indonesian Gated Onboard Robotics



iSeeDesignITB

AI/Edge Accelerator focus (Topic 3)

PERURI Chip Hackathon 2026

20th October 2026

PERURI's fifth phase puts sensors in factories

1
Banknotes
Cash people can trust

2
Security printing
Documents with security built in

3
Digital security
PP 6/2019: e-signatures, e-Meterai

4
GovTech
SPBE: government apps, coordinated by PERURI

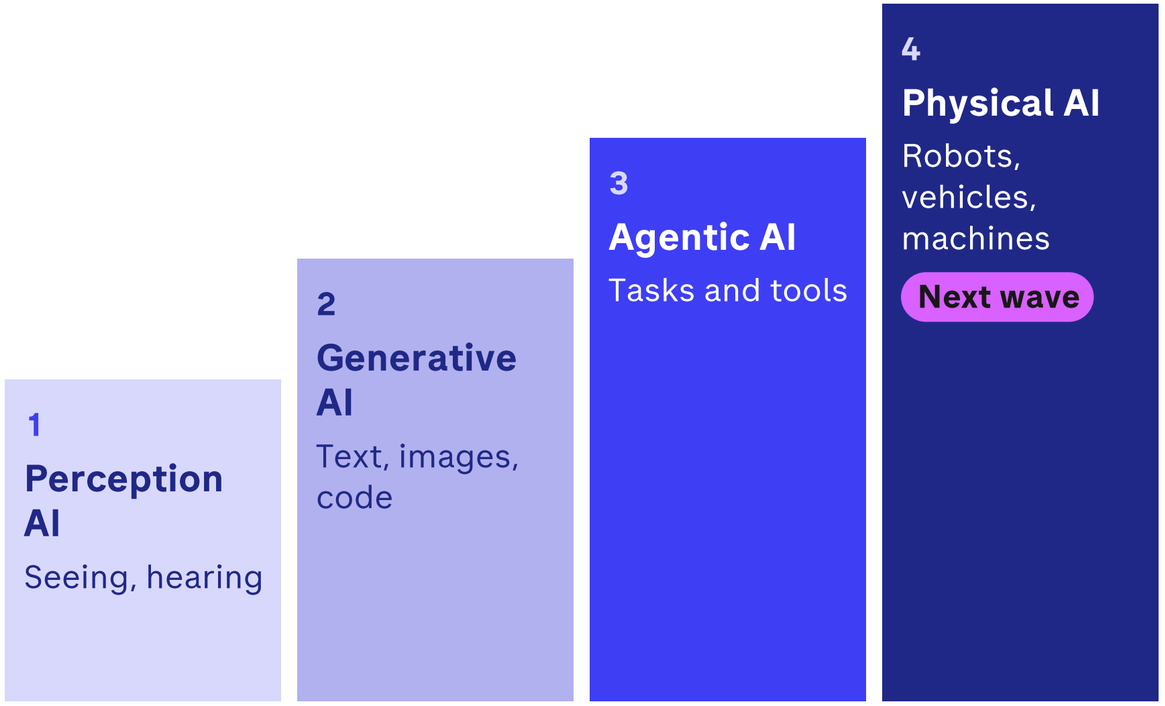
5 · 2030
Security tech
Sensors in every excise (cukai) factory

The chip: not yet

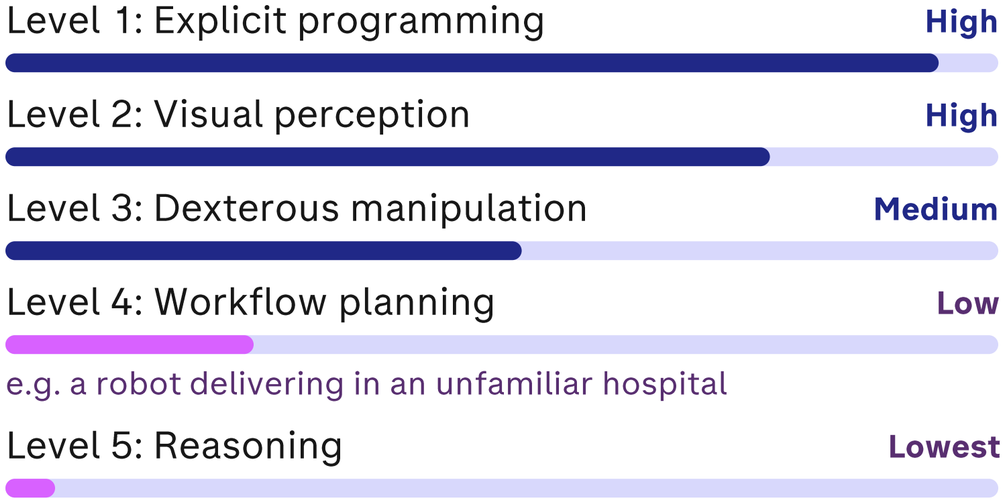
Sensors will no longer be static.
They're going to move with robots.

AI is leaving the screen for the physical world

Four waves of AI, as NVIDIA describes them



BCG: five levels of physical AI, by maturity

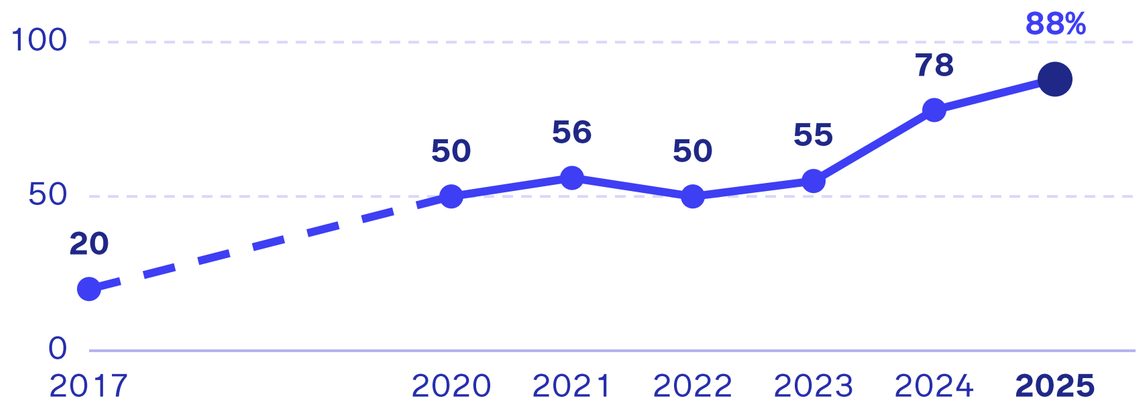


Moravec's paradox: what is easy for people, like crossing a cluttered aisle, is hard for machines.

In a robot, applied AI's first job is to avoid people

Organizations using AI in at least one function, %

McKinsey global survey. The definition of "use" changed over time.



87% of manufacturing AI users cut costs; 63% grew revenue

When AI moves a machine, a wrong output is a collision

41 robot-related deaths at US workplaces, 1992–2017; 78% were a robot striking a worker (NIOSH)

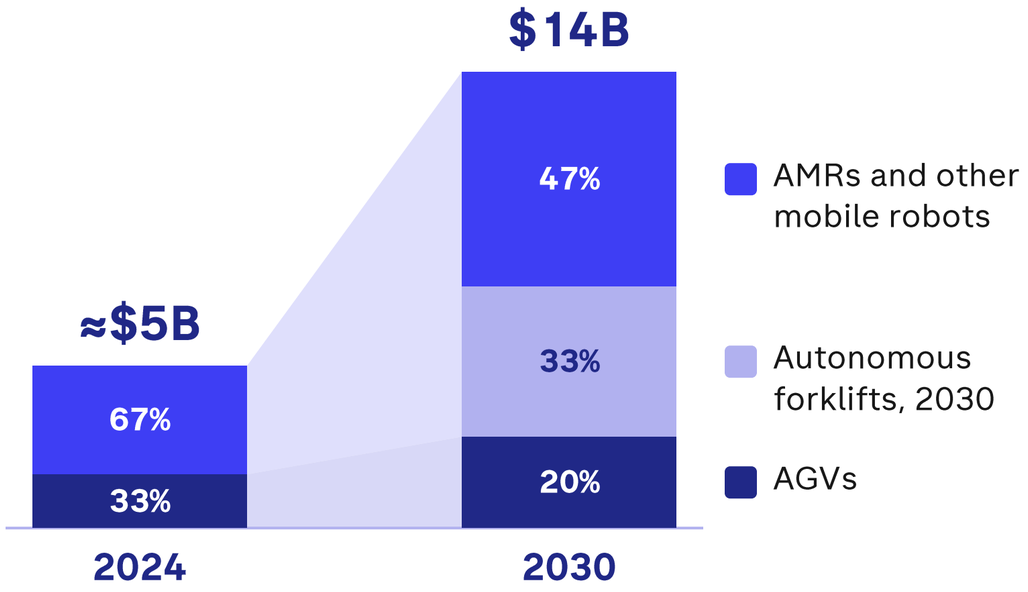
85 US deaths a year from forklifts alone, plus 34,900 serious injuries (OSHA)

Mobile robots take away the fence. The limit has to live inside the machine.

Mobile robots are scaling; cash handling is automating too

Global mobile robot revenue, Interact Analysis

AGVs fall from 33% to 20% of revenue as AMRs take over



Central Bank of Egypt

The same business as PERURI: printing and handling banknotes

Banknote print house → **Highly automated cash center**

Egypt's central bank runs a highly automated cash center, linked to its print house (Giesecke+Devrient).

PERURI prints Indonesia's money. High-value flows like these are where automation goes next.

Indonesia has come late to every tech wave



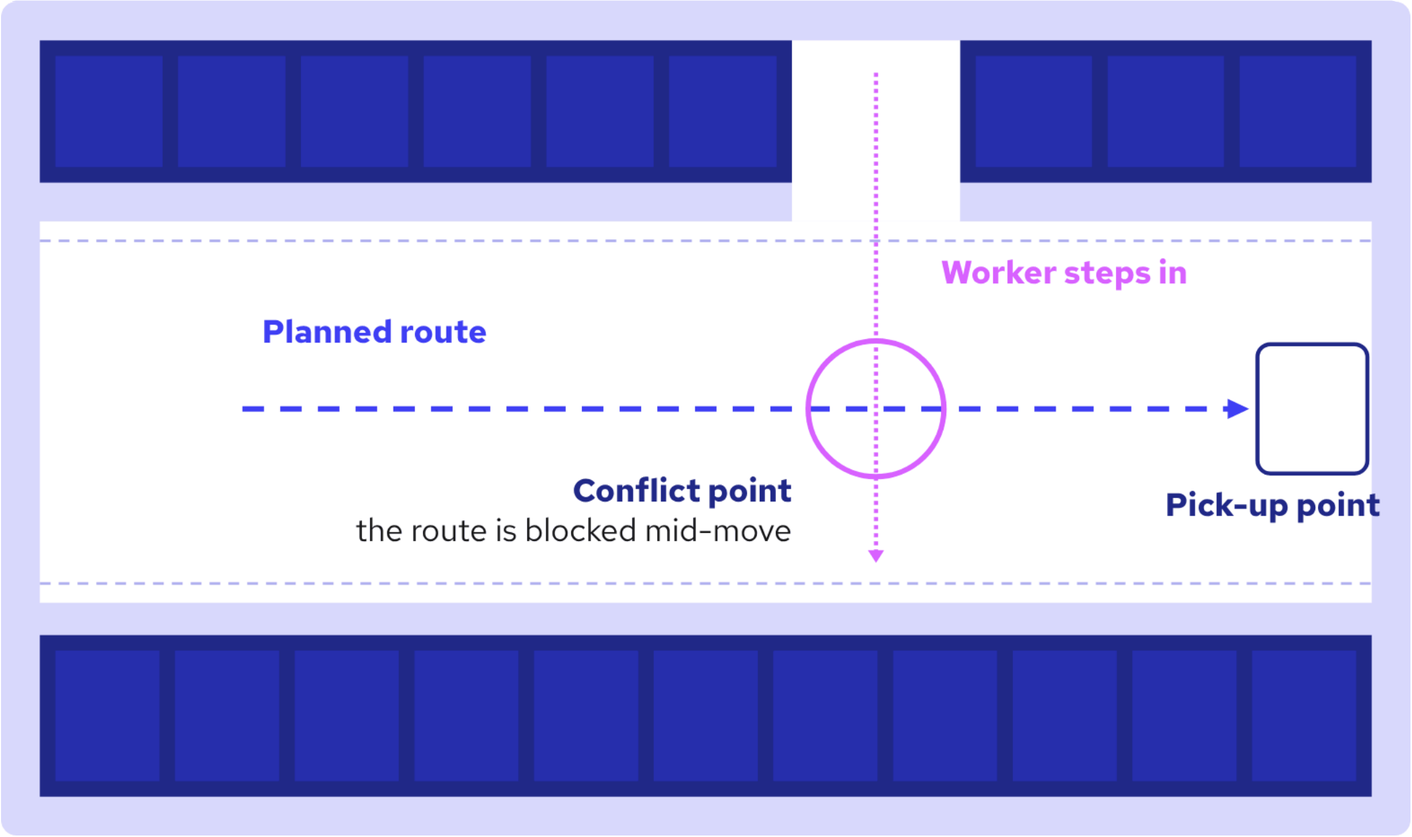
15.51B transactions
worth Rp1,420.66T in 2025; 59.98M users (Jan 2026)

US$90B digital economy
2024 GMV, the largest in Southeast Asia

10× nickel exports
US$3.3B (2017) to US$33.5B; EV sales 4.2× in 2022–24

We came late to QR payments, 4G and EVs, and caught up only once the state built the base. Physical AI is the next wave: let's not be late again.

The gap IGOR fills: Human–Robot Coexistence



Illustrative operating scenario, based on the crossing-pedestrian and narrow-passage cases in our training curriculum.

What happens in the aisle

1 Moving

The robot follows its planned route through a shared aisle.

2 Interrupted

A worker steps into that route.

3 Decide, every 20 ms

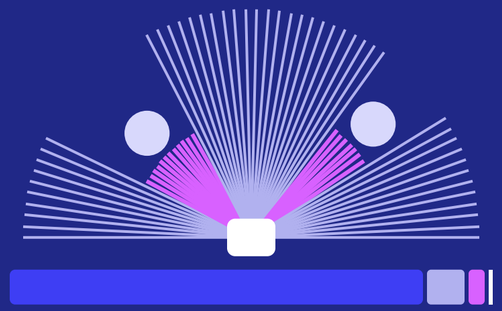
The robot has to adapt right away, without breaking its limits.

IGOR lets the robot adapt, and keeps the AI's correction inside a fixed range.

Control loop

1 Sense

74 inputs, every 20 ms

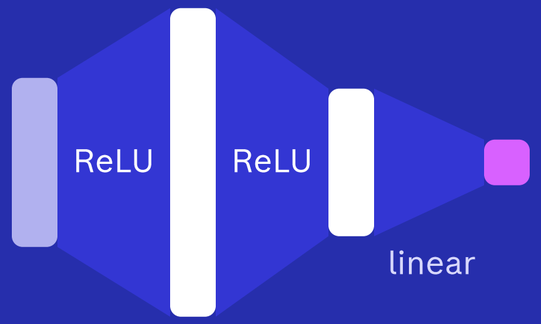


64 LiDAR, 6 IMU,
3 command, 1 heading

Nearest hit per sector keeps thin obstacles visible; every input is scaled to −1…+1.

2 Infer, on the FPGA

74 → 128 → 64 → 3 network



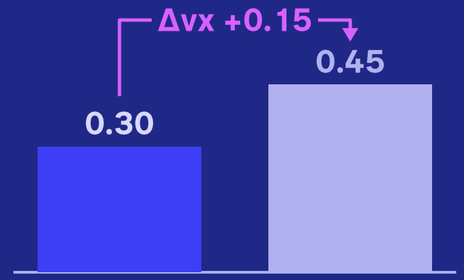
138.32 µs

a fixed 6,916-cycle schedule: 17,856 MACs in Q8.8 across 8 parallel lanes, bit-exact to the trained model

3 Bound

3 corrections, capped in hardware

Example forward speed, m/s



Δvx −0.60 to +0.15 m/s

Δvy ±0.45 m/s

Δω ±0.80 rad/s

Braking authority is 4× the forward boost.

4 Act

Host and STM32

Blend with the tracker by α

CBF safety filter

Speed envelope, stop at 0.22 m

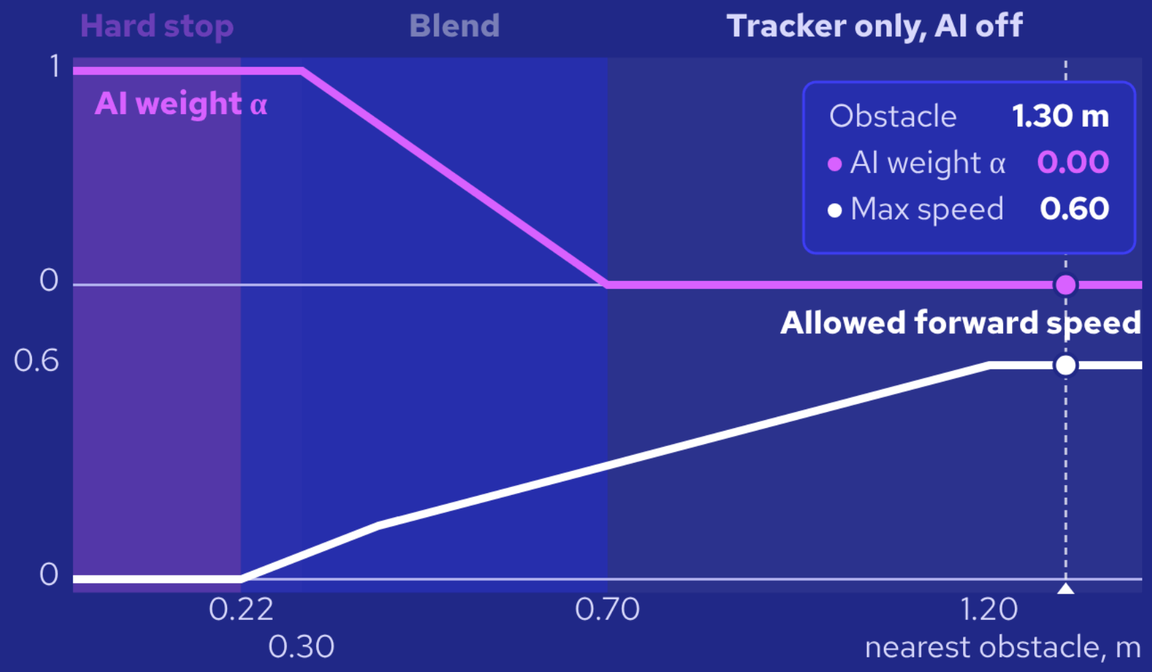
Mecanum kinematics

PID on all four wheels

The tracker drives. The network only corrects, and the hardware caps how far.

Gating and training

How much the AI may correct, by distance to the nearest obstacle



Curriculum PPO training, 300k steps

1	2	3	4

1 **Static obstacles, 0–20% of steps**
Corridor tracking, smooth braking

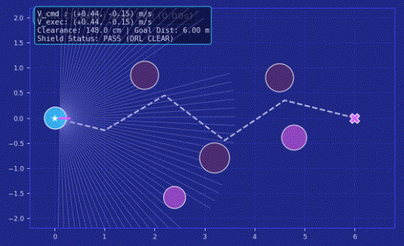
2 **Crossing pedestrians, 20–45%**
Reads obstacle motion from beam changes

3 **75 cm choke points, 45–70%**
15 cm per side, without freezing

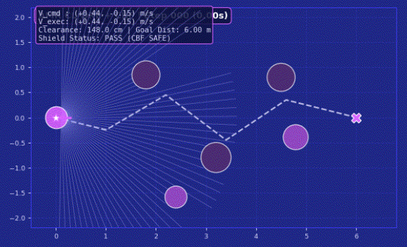
4 **Chaos and randomization, 70–100%**
4 moving obstacles, wheel slip, noise and lag

PPO, FP32 → Q8.8 bit-exact → FPGA ROM

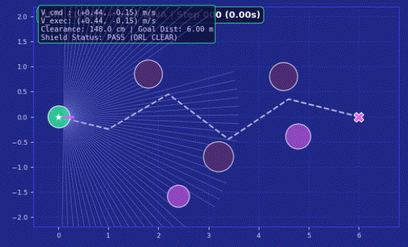
Baseline runs



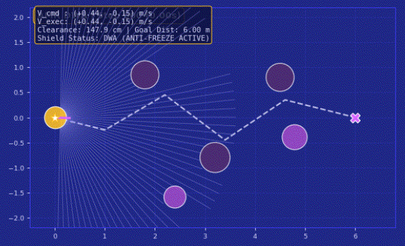
CL-DRL alone
Learned policy only



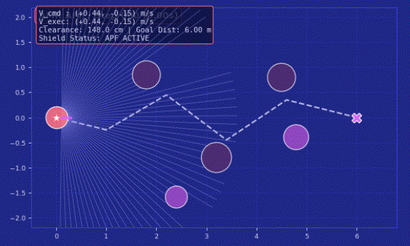
CL-DRL + CBF (ours)
Learned policy + safety filter



CL-DRL + mini-DWA shield
Learned policy + rollout shield

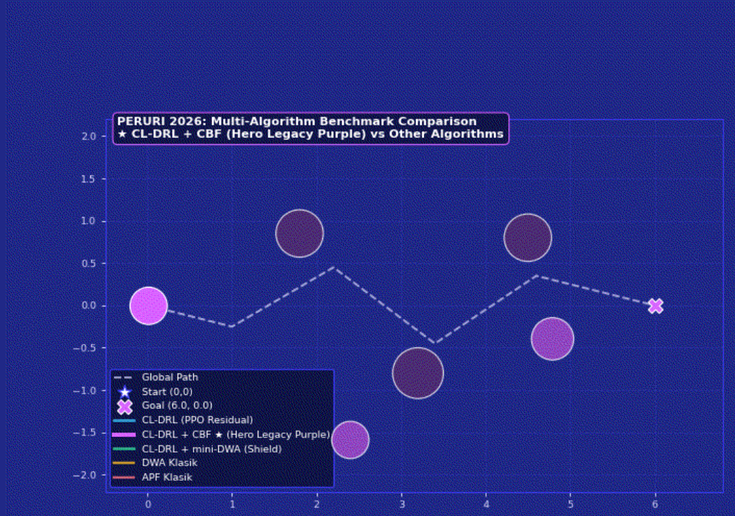


Classical DWA
Dynamic window, no learning



Classical APF
Potential field, no learning

Head-to-Head comparison



All five controllers, same start, same obstacles, same 50 Hz loop

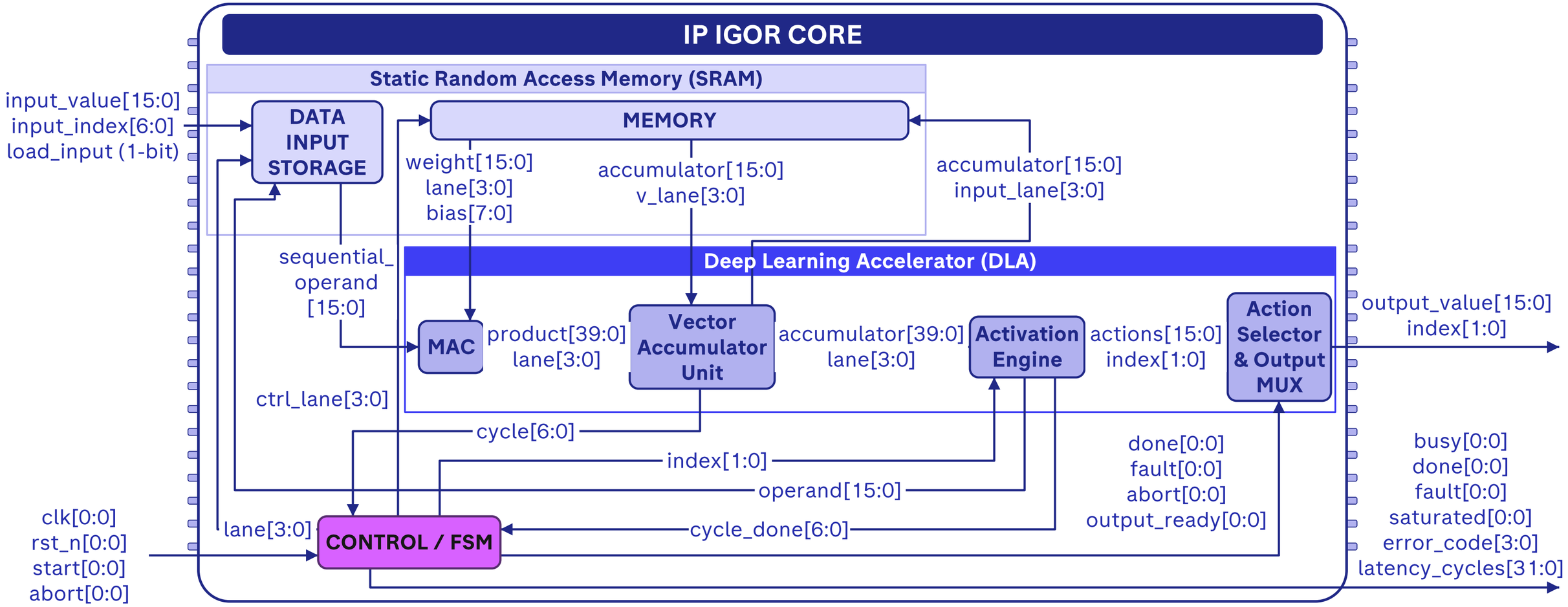


Level 4 chaos course	Collisions	Freezing	m/s
CL-DRL alone	<3%	0%	0.508
CL-DRL + CBF (ours)	0%	0%	0.502
CL-DRL + mini-DWA shield	0%	<2%	0.485
Classical DWA	0%	5–10%	0.380
Classical APF	15–20%	25%	0.420

Simulation benchmark from the training pipeline, not a field result.

Only CL-DRL + CBF reaches 0% collisions and 0% freezing while cruising above 0.5 m/s.

Datapath


IP IGOR CORE
Static Random Access Memory (SRAM)
DATA INPUT STORAGE
MEMORY
Deep Learning Accelerator (DLA)
MAC
Vector Accumulator Unit
Activation Engine
Action Selector & Output MUX
CONTROL / FSM
input_value[15:0]
input_index[6:0]
load_input (1-bit)
weight[15:0]
lane[3:0]
bias[7:0]
accumulator[15:0]
v_lane[3:0]
accumulator[15:0]
input_lane[3:0]
sequential_ operand [15:0]
product[39:0]
lane[3:0]
accumulator[39:0]
lane[3:0]
actions[15:0]
index[1:0]
output_value[15:0]
index[1:0]
ctrl_lane[3:0]
cycle[6:0]
index[1:0]
operand[15:0]
done[0:0]
fault[0:0]
abort[0:0]
output_ready[0:0]
busy[0:0]
done[0:0]
fault[0:0]
saturated[0:0]
error_code[3:0]
latency_cycles[31:0]
clk[0:0]
rst_n[0:0]
start[0:0]
abort[0:0]
lane[3:0]
cycle_done[6:0]

01 / IGOR RTL dataflow

MODEL-SPECIFIC DATAFLOW

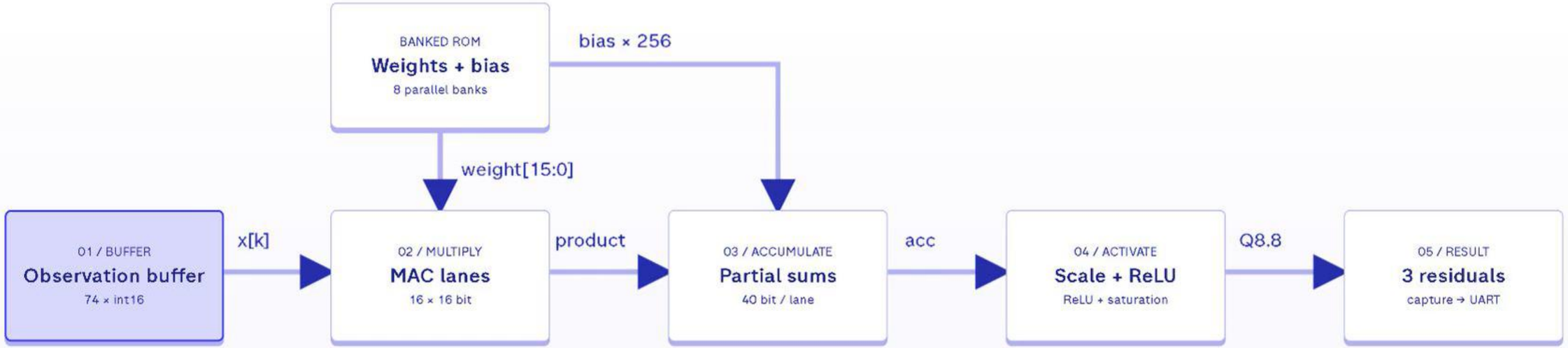
ANIMATED DATAFLOW / RTL SCHEDULE MODEL

One inference, from input to residuals

Educational playback of the checked schedule. The display is slowed down for explanation.

0 / 6,916

clock core / 50 MHz reference



IDLE

Input ready

A complete observation is available before the core starts.

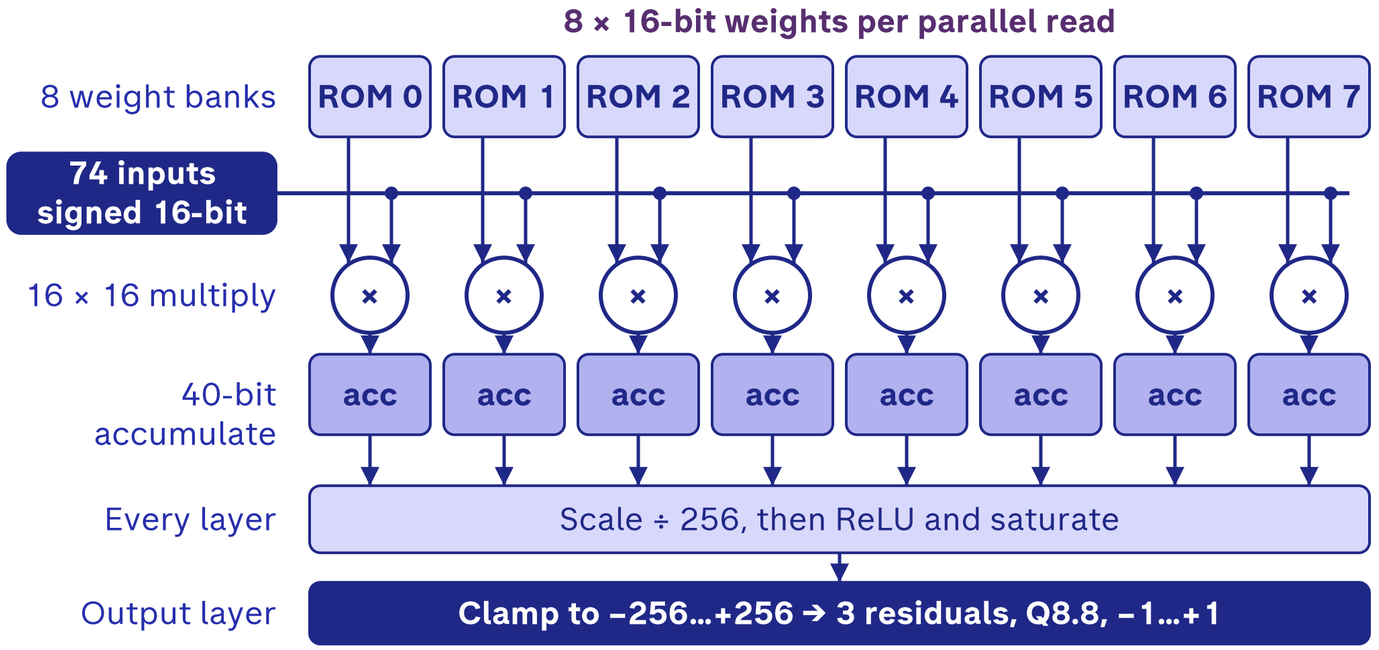
start → BIAS_ADDRESS

Layer / group	Operand / neuron
FC 1 / 1 / 16	neuron 0
Active lanes	**Core status**
8 / 8	idle

One complete inference

74 normalized inputs enter a fixed neural schedule.

Parallel Lanes



Share of the Cyclone V used by the fitted design

ALMs needed	3,303 of 41,910, 7.9%
M10K blocks	64 of 553, 11.6%
DSP blocks	8 of 112, 7.1%
FPGA PLLs	0 of 6, 0%

Each row has its own denominator; they don't add up. Lowers area usage, latency, and throughput

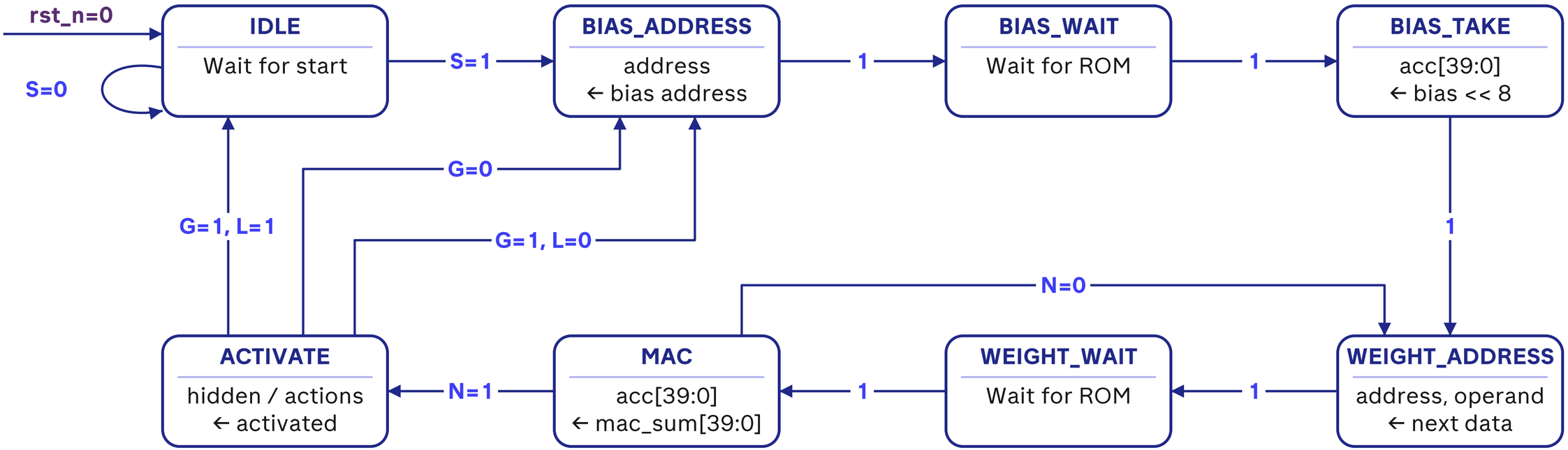
Output layer: 3 lanes used, 5 idle



97.5% of neuron slots useful: 195 of 200 across 25 groups of eight

35 KiB of weights and biases, padded to a 64 KiB on-chip ROM

Core FSM



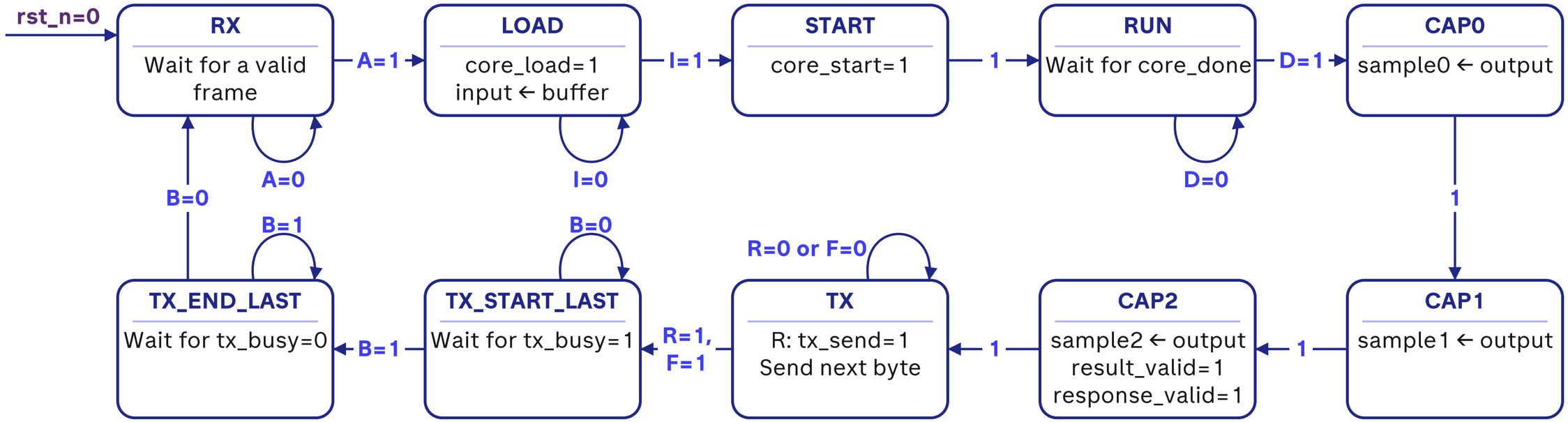
S=start

N=(k==input_count-1)

G=(neuron+8>=output_count)

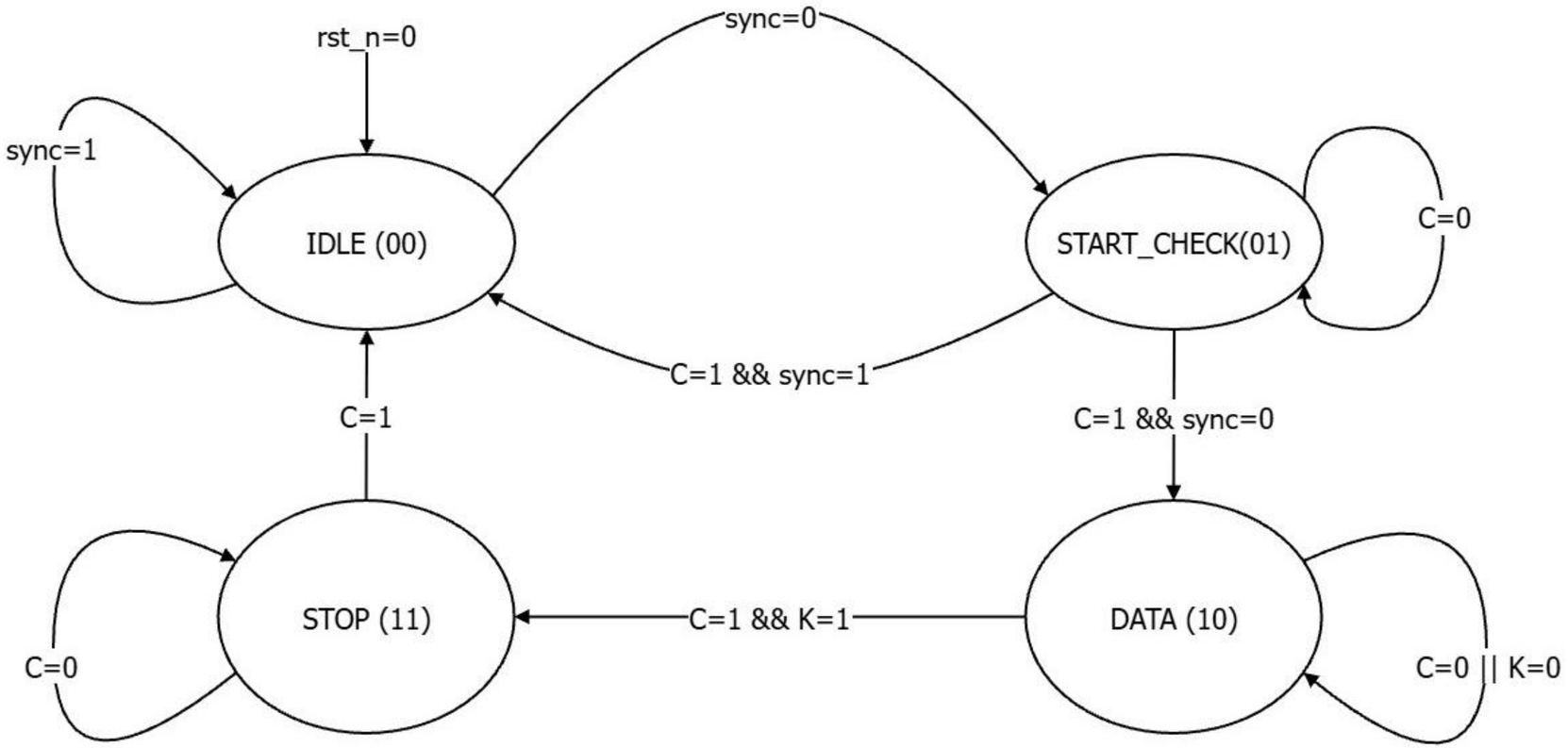
L=(layer==2).

Controller UART FSM



A=Check Checksum sah
I= 74 Input sudah masuk
D=core_done
B=tx_busy.
R=!(tx_busy || tx_send)
F=(en_tx).
Saat R=1 && F=0: tx_index++.

Receiver FSM



C = Check apakah bit yang dicari ditemukan
Sync = Tombol Sync untuk set UART
K = Parity Bit

Functional FPGA Coverage

DE10-Nano board top, real 50 MHz and 921,600 baud

18 checkpoints: reset, KEY1 stop, KEY0 recovery, an independently decoded reply.

UART transaction controller + core

184 directed transactions, 268 assertions, all 74 input-field boundaries, CRC, framing and recovery

Neural core, 8 lanes

236 network vectors, every one of 192 hidden activations and 3 outputs compared; 8,364 activation edge cases; 54,399 assertions

Safety gate unit

111,210 oracle comparisons across all 12 outcomes

Jetson host software

29 of 29 regression tests; LiDAR driver 16 of 16

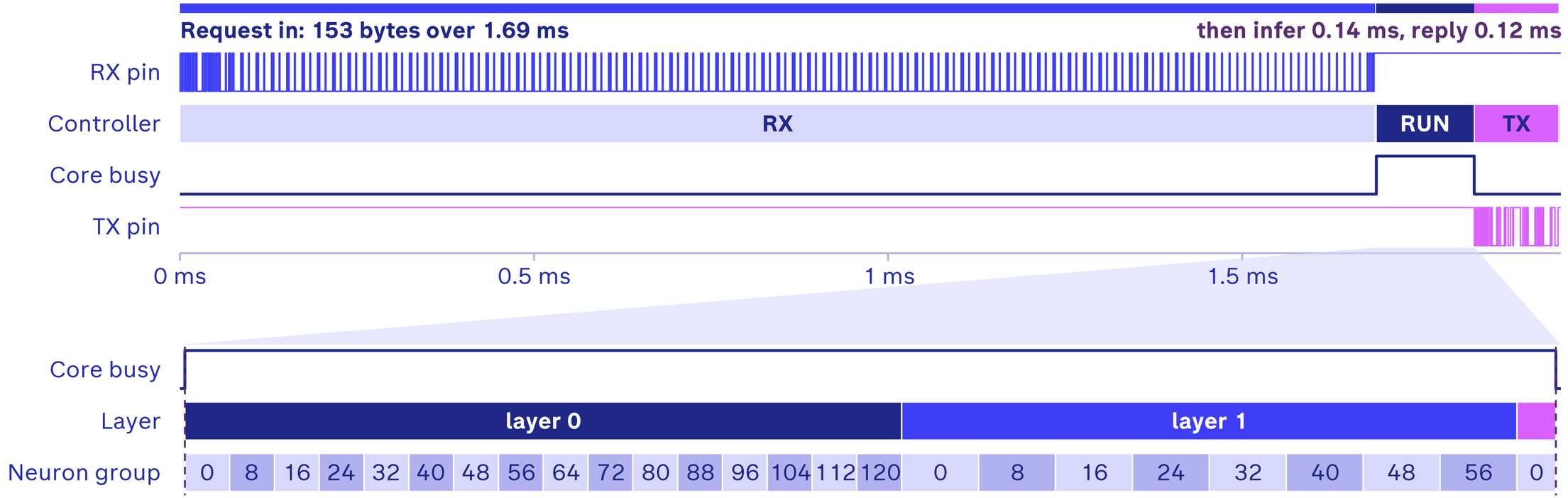
Functional bins reached

8 / 8	core FSM states
10 / 10	core FSM normal transitions
22 / 22	safety-gate decision outcomes, 111,210 vectors
74 / 74	input-field boundaries, through the UART

The evidence is checked too: 29 waveform views re-read from the simulation dump, 18,176 MAC updates checked lane by lane, 0 changed hashes across sources, weights and bitstream.

RTL simulation in Verilator and Icarus Verilog, not board measurement.

Clock Cycle by Design



16×(3+74×3+1) + 8×(3+128×3+1) + (3+64×3+1) = 6,916 clocks = 138.32 µs between the markers;
16 + 8 + 1 groups of eight neurons

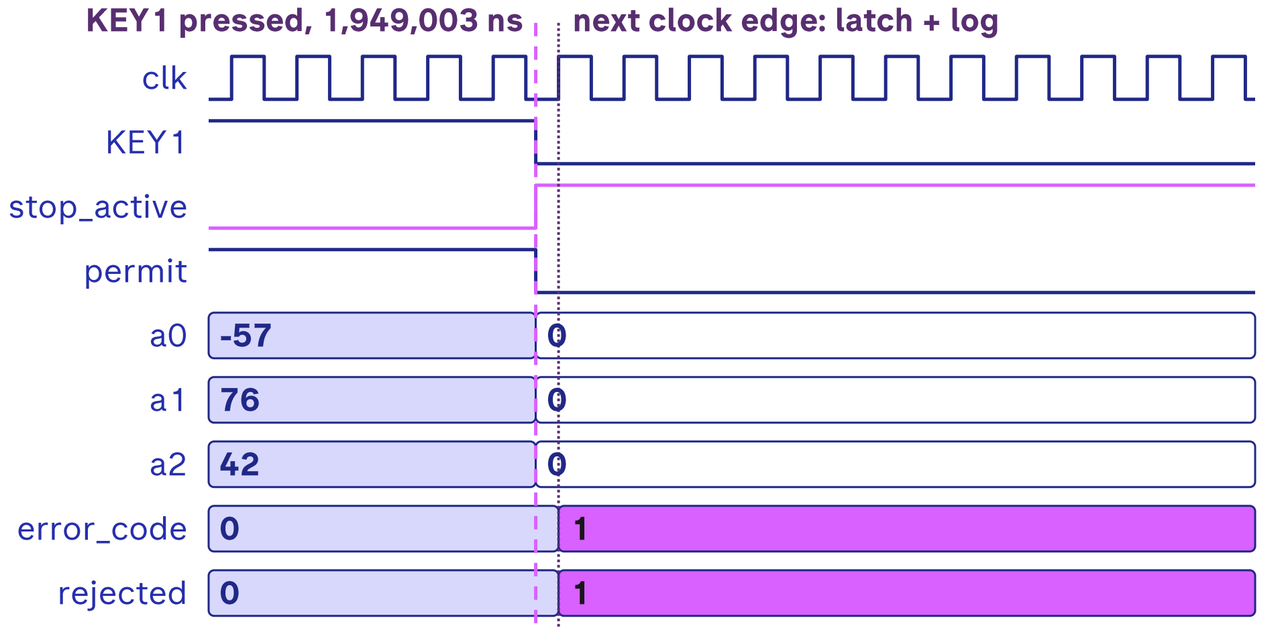
Reply on the wire, golden vector 1

5A A5	02 06	C7 FF	4C 00	2A 00	0F
header	reply, 6 bytes	−57 = −0.223	76 = +0.297	42 = +0.164	CRC-8

Throughput Clock_freq/Clock_cycle = 7229.61249 Output/s

Fault Coverage

Emergency stop, from the board simulation



The residuals drop in the same instant KEY1 falls, without waiting for a clock. Releasing KEY1 does not restart; only KEY0 does.

8 / 8 injected fault classes end in zero correction

1	KEY1 emergency stop
2	Low stop bit, framing
6	Corrupted CRC
7	Partial frame stalls
9	Byte arrives during inference
10	Core reports a fault
11	Result older than its lease
12	Core never finishes, watchdog

Each one: error code logged, transaction rejected, no reply sent, all three residuals held at 0

Code Coverage

RTL module, suite	Line	Branch	Toggle
Neural core core suite	**100%** 68 of 68	**100%** 96 of 96	**95.8%** 870 of 908
UART controller UART suite	**97.9%** 46 of 47	**100%** 46 of 46	**71.8%** 371 of 517
UART receiver UART suite	**90.0%** 9 of 10	**100%** 14 of 14	**70.5%** 31 of 44
UART transmitter UART suite	**100%** 5 of 5	**100%** 4 of 4	**69.8%** 30 of 43
Board top board suite	**100%** 4 of 4	**100%** 6 of 6	**26.3%** 51 of 194
Safety gate gate suite	**100%** 13 of 13	not measured	**95.3%** 102 of 107

100%

line and branch coverage of the neural core, raw counters, nothing waived

Why toggle is lower

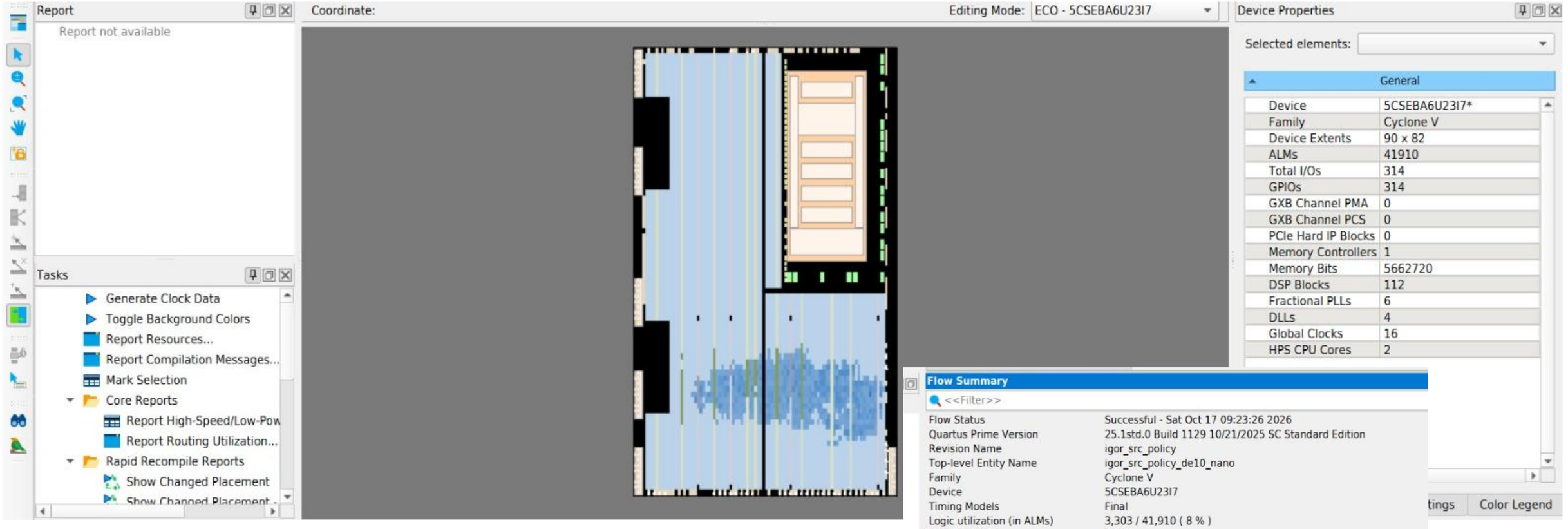
Most missing toggles are bits that never change in legal use, like the top bits of a latency counter that always reads 6,916. They stay in the denominator.

Coverage found a missed line

The illegal-state default was never hit. We injected state 15, checked a fail-safe error 6, and line coverage went from 67 to 68 of 68.

Standar Code Coverages (IEEE Standard for SystemVerilog– Unified Hardware Design, Specification, and Verification Language 2017) : https://rfsoc.mit.edu/6S965/_static/F24/documentation/1800-2017.pdf

Chip Design Estimation, Timing No Issue, Only 8% Logic Utilization



Multicorner Timing Analysis Summary

<<Filter>>

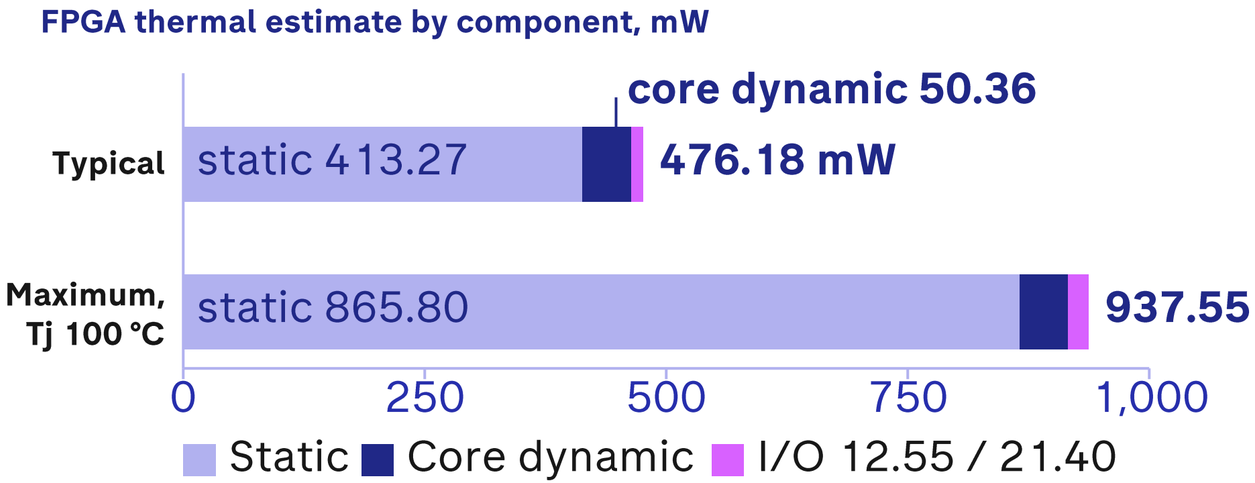
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	Worst-case Slack	5.647	0.119	14.279	0.253	8.462
1	FPGA_CLK1_50	5.647	0.119	14.279	0.253	8.462
2	Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	FPGA_CLK1_50	0.000	0.000	0.000	0.000	0.000

Flow Summary

<<Filter>>

Flow Status	Successful - Sat Oct 17 09:23:26 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Standard Edition
Revision Name	igor_src_policy
Top-level Entity Name	igor_src_policy_de10_nano
Family	Cyclone V
Device	5CSEBA6U23I7
Timing Models	Final
Logic utilization (in ALMs)	3,303 / 41,910 (8 %)
Total registers	6277
Total pins	13 / 314 (4 %)
Total virtual pins	0
Total block memory bits	524,288 / 5,662,720 (9 %)
Total DSP Blocks	8 / 112 (7 %)
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0 / 6 (0 %)
Total DLLs	0 / 4 (0 %)

FPGA Power Coverage and Memory Utilization



M10K blocks	64 / 553	12 %
Total MLAB memory bits	0	
Total block memory bits	524,288 / 5,662,720	9 %
Total block memory implementation bits	655,360 / 5,662,720	12 %
Total DSP Blocks	8 / 112	7 %
Fractional PLLs	0 / 6	0 %
Global signals	1	

50.36 mW ULP Chip

02 / Verification and recovery

BROWSER MODEL / NO RTL SIMULATOR CONNECTED

TINYTAPEOUT / PROTOCOL WALKTHROUGH

Load. Commit. Infer.

Explore transaction preconditions in a simplified browser model. It does not compute neural outputs or replace cocotb.

● WAITING / complete the next step — BROWSER MODEL

0%	0 / 6,020	—
Workflow progress	Parameter words staged	Reference cycles / not elapsed clocks

○ Image loaded ○ Inputs loaded ○ Result available

Empty | Loading | Ready | Inputs loaded | Done

BEGIN | Upload 6,020 words | COMMIT | Load 24 inputs | START → | Reset

Command activity

RESET
Accepted

Phase: EMPTY
Uploaded words: 0 / 6020
Awaiting command…

NUMERIC INSPECTOR

Signed Q8.8

Inspect how a value maps to a signed 16-bit word. Rounding here uses nearest with ties away from zero.

Real value (−128 to 127.99609375)

0,5

● Live numeric view — Q8.8

Drag to inspect a value

−128 — 0 — +127.996

Positive / decoded 0.5 / raw 128

0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Sign bit — Integer bits / fractional bits — LSB

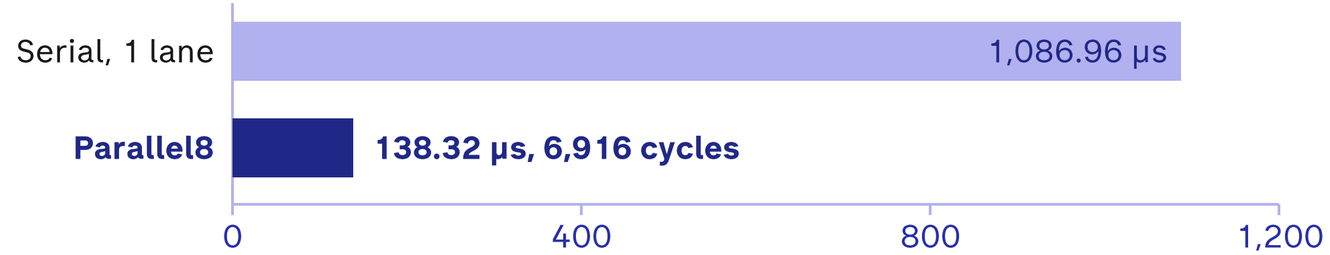
Signed integer 128
Hex word 0×0080
Decoded value 0.5
Error 0.00000000

A transaction has preconditions

This browser model illustrates the TinyTapeout neural protocol.

7.86x speedup due to Parallelism Technique compared by calculation

Neural kernel latency in µs, same 74→128→64→3 policy at 50 MHz



−87% kernel time; 98% of an ideal 8× speedup

Tradeoff: **+54 ALMs (+1.7%), +281 registers**

DSP blocks **1 → 8**. M10K unchanged at 64.

50 MHz
core clock, straight from the board

0 PLLs
of 6 on the FPGA used

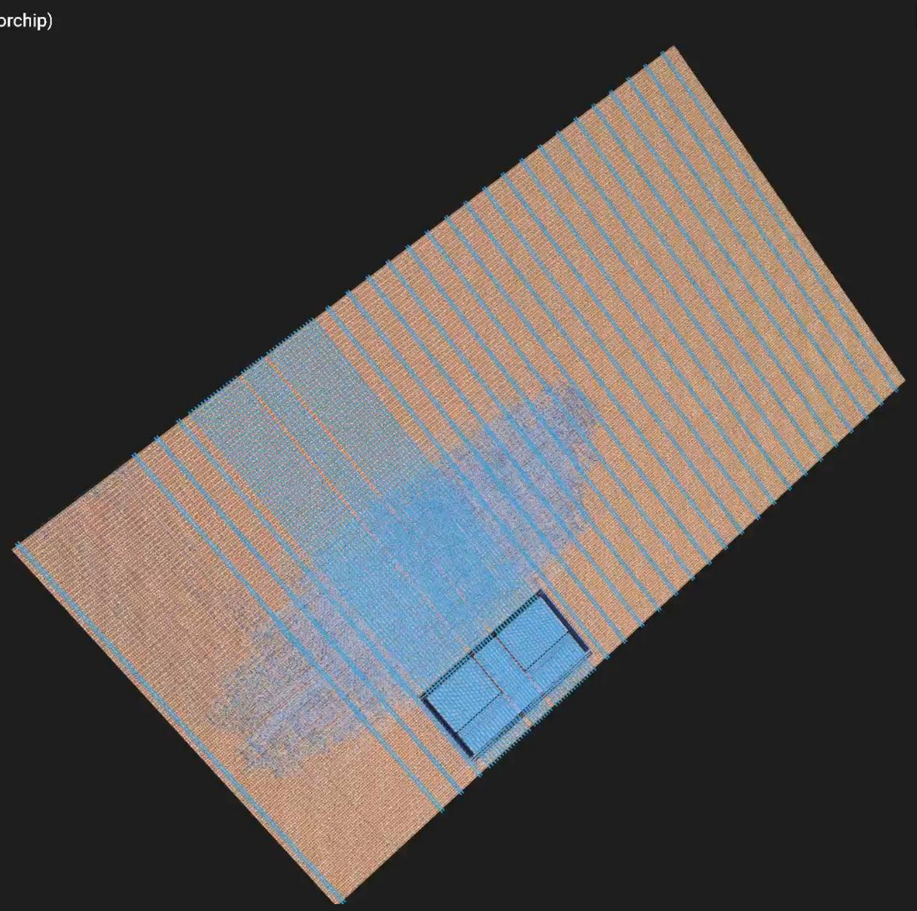
Setup slack, worst case	**+5.647 ns**
Hold slack, worst case	**+0.119 ns**
Total negative slack	**0 ns**

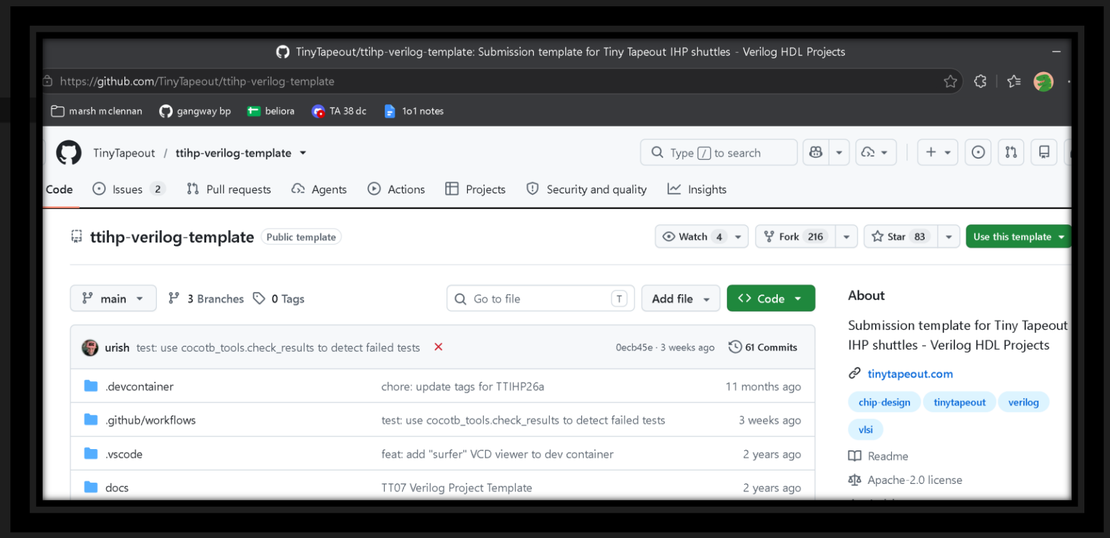
Reported multicorner checks pass at 50 MHz.

98/98 physical Jetson ⇄ FPGA UART requests correct; round-trip time in ms



GDSII Layout





The GDSII layout of the IGOR chip

Built with the Tiny Tapeout flow on the IHP open-source PDK: every metal layer and standard cell the foundry would print. No DRC Error, No LVS Error, No Antenna Violation, No Timing Violation. Default ASIC clock frequency for proof of concept.

Area analysis

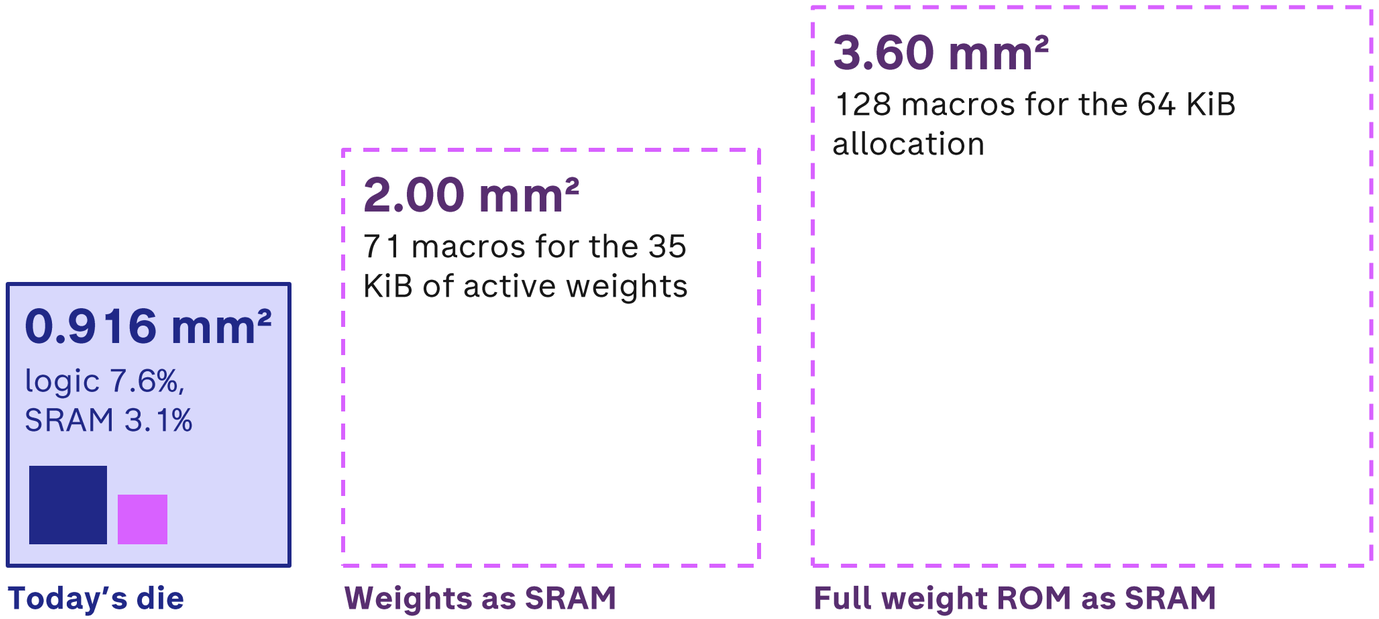
Where the 3,303 FPGA ALMs go (fitted hierarchy)



Neural core 2,402.7 ALMs, 72.7%

UART controller 838.2, RX 38.3, TX 21.5, top 2.3

Silicon area, drawn to scale (I/O chip, IHP SG13CMOS5L)



Today's die | **Weights as SRAM** | **Full weight ROM as SRAM**

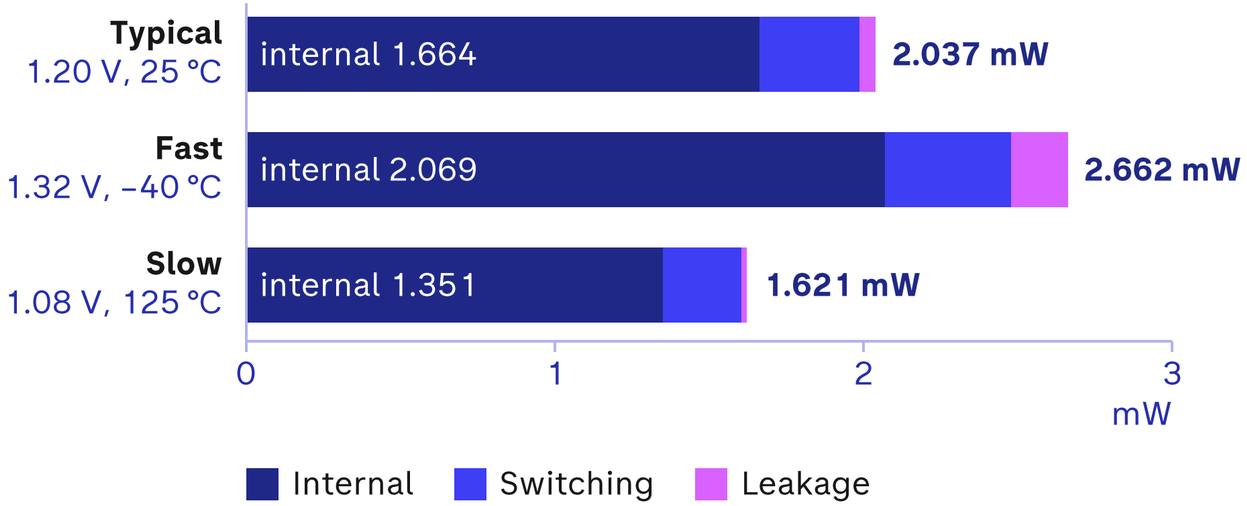
2.2–3.9×

today's whole die, for the weights alone in this SRAM macro

Next on silicon: 9-bit coefficients and a dedicated weight ROM. Memory-only projection; logic, routing and power networks excluded.

Power analysis

Post-route power by corner, tt_um_igorchip on IHP SG13CMOS5L



0.916 mm²
die area, with one 256 × 16-bit program SRAM

4
protocol contexts with separate state, pin ownership in hardware

PASS
gate-level test of the routed netlist, 74,821 ns simulated

Typical 2.037 mW by group: where the power goes



Security by layer: what holds today, what we close next

Layer	In place today	Next
FPGA neural core DE10-Nano, Cyclone V	E-stop zeroes outputs within one clock; all 74 inputs range-checked; 2 ms inference watchdog; results expire after 20 ms	Wire the speed and rate-limit gate into the active bitstream; encrypt the bitstream; hash-check the weights
⚠ **UART 921.6 kbps, CRC-8**	No authentication: a frame can be forged or replayed	Sequence numbers now; HMAC via a secure element later
Host, ROS 2 Jetson Orin Nano	Speed envelope with a hard stop at 0.22 m; command and sensor leases stop the robot when data goes stale	Feed live LiDAR obstacles to the CBF filter; fail closed if the FPGA link drops; SROS2 access control; out-of-distribution input check
Motor controller STM32, USB serial 1 Mbps	Watchdog cuts motor PWM when commands stop arriving	Scale all four wheels together so heading holds at speed limits
I/O engine ASIC IHP shuttle, not in the robot loop yet	Pin ownership per context; program writable only while the core is idle; 42 tests and gate-level pass	Hardware root of trust before any production spin

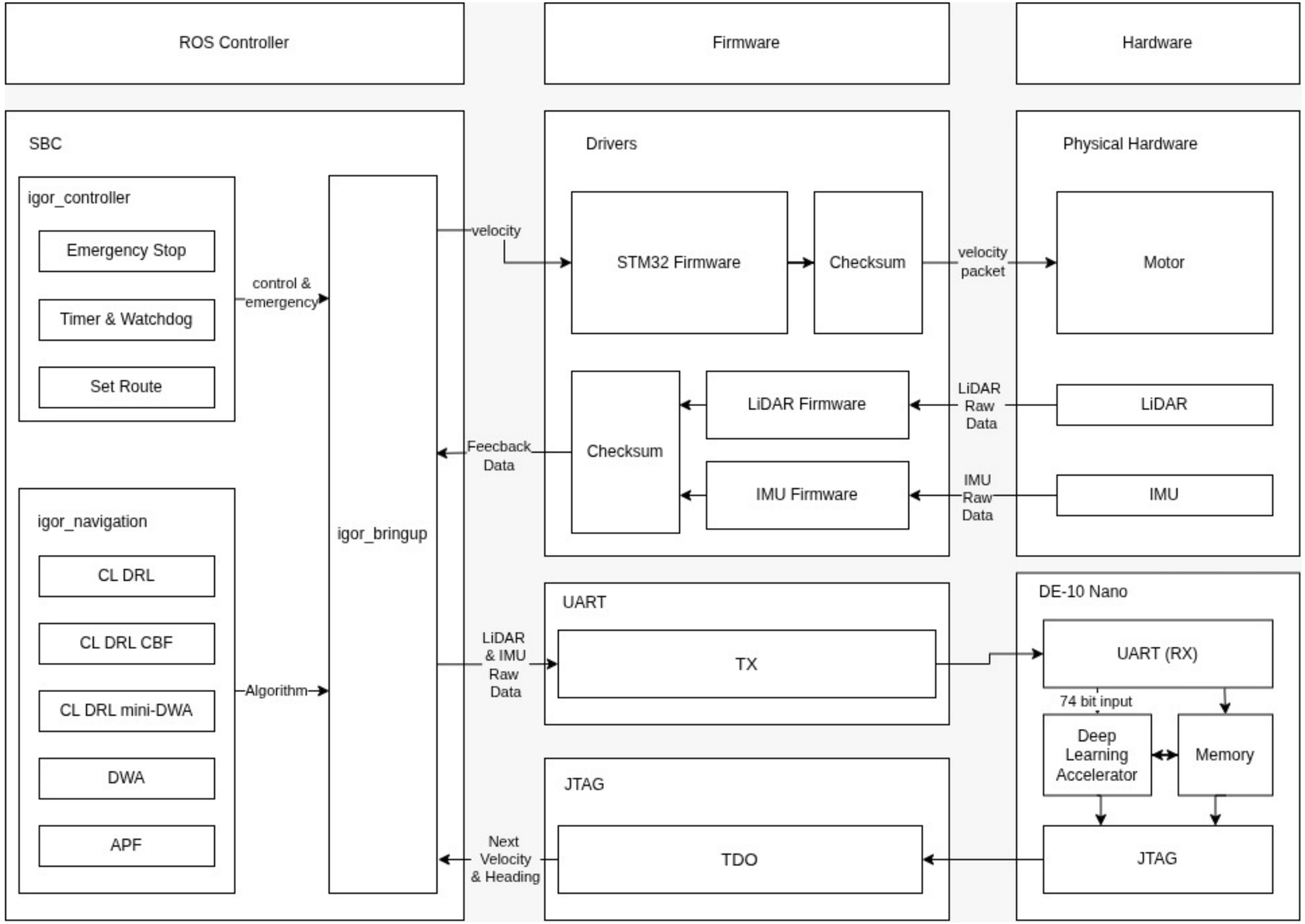
Self-audit against ISO/IEC FDIS 27090 (AI security), Oct 2026

6 pass	6 critical	6 high	4 medium

Every finding has a fix and an owner

One release manifest hashes RTL, weights, bitstream and host constants; CI tests the code that ships.

System Architecture



18 intelligent disturbances: 11 verified, 7 closing next

Status	Disturbance	IGOR response
People and cargo		
Sim	**Worker steps in close**	Hard stop under 0.22 m
RTL	**Operator hits E-stop**	FPGA zeroes motors, logs a fault
RTL	**AI surges near cargo**	Forward capped at 0.15 m/s
Next	**Load shifts in a turn**	Bounded fixes; OOD check next
Environment and power		
RTL	**Facility blackout**	LiDAR works dark; stale map stops
RTL	**Liquid on the wiring**	Corrupt frames rejected
Sim	**Wet, oily floor**	50 Hz closed loop + wheel PID
Next	**Battery brownout**	STM32 cuts PWM after 200 ms
RTL	**Press-line EMI**	CRC-8 drops corrupted frames

Status	Disturbance	IGOR response
System and hardware		
RTL	**Host OS jitter**	Fixed 138.32 µs on the FPGA
Sim	**Serial latency spike**	Trained with 5–15 ms lag
Next	**Sensor hangs the I²C bus**	Each I/O context isolated
Next	**Two drivers, one pin**	Pin mask rejects and logs it
Cyber attack and supply chain		
RTL	**LiDAR or IMU spoofing**	Outputs clamped to ±1.0 in RTL
Next	**Serial frame replay**	Sequence numbers + HMAC next
RTL	**Weight tampering**	Weights in ROM, no write path
Next	**Hacked host rewrites I/O**	Microcode locked while running
Next	**Tampered bitstream**	Hash manifest; encryption next

RTL **8** in RTL or gate-level sim Sim **3** in system tests or simulation Next **7** mitigated, closing next 111,210 RTL test vectors

PERURI's fifth phase needs a chip it doesn't have yet

…and EU law is creating the demand: a regulated market, a sovereign chip line, an internal first customer

1 Banknotes	2 Security printing	3 Digital security	4 GovTech	5 Security tech, 2030
Cash people trust	Documents built secure	e-signatures, e-Meterai	SPBE state apps	IGOR: sensors that move

Why it fits PERURI

Phase 5, 2030
Sensors in every cukai factory, and they will move with robots. The chip line is still "not yet": IGOR is its first product [1]

Same security DNA as phases 1–4
Weights in ROM with no write path, outputs clamped in RTL; 11 of 18 factory and cyber disturbances verified, 7 closing next [2]

Sovereignty
Chip imports hit US$4.87bn in Jan–Nov 2025 [3]. IGOR's fab path, US$8,000 per 1,000 dies, fits PERURI's chip sandbox and Danantara's planned Arm partnership [4] [5]

First customer is PERURI itself
A paid FPGA pilot at a PERURI site in 2027

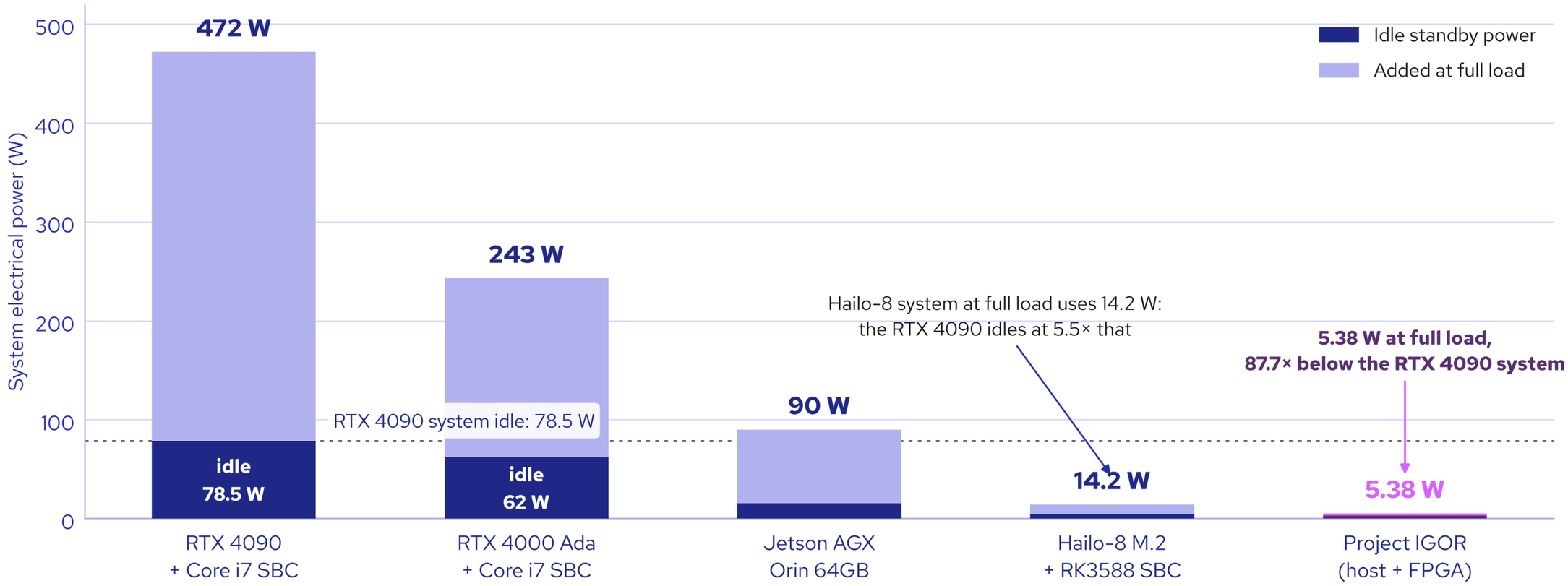
Demand is locked in

20 Jan 2027	EU Machinery Regulation 2023/1230: AI safety parts need a third-party exam [6]
2 Aug 2028	EU AI Act duties start for AI in machinery [7]
2030	279k new mobile robots a year, 151k outside China; 4.2M installed [8] [9] [10]
IGOR target	2,000 robots a year in 2030, 1.3% of the market we can sell into; 11,000 in 2036, 3.7%

So what for PERURI: a regulated, growing market, a sovereign chip line and an internal first customer

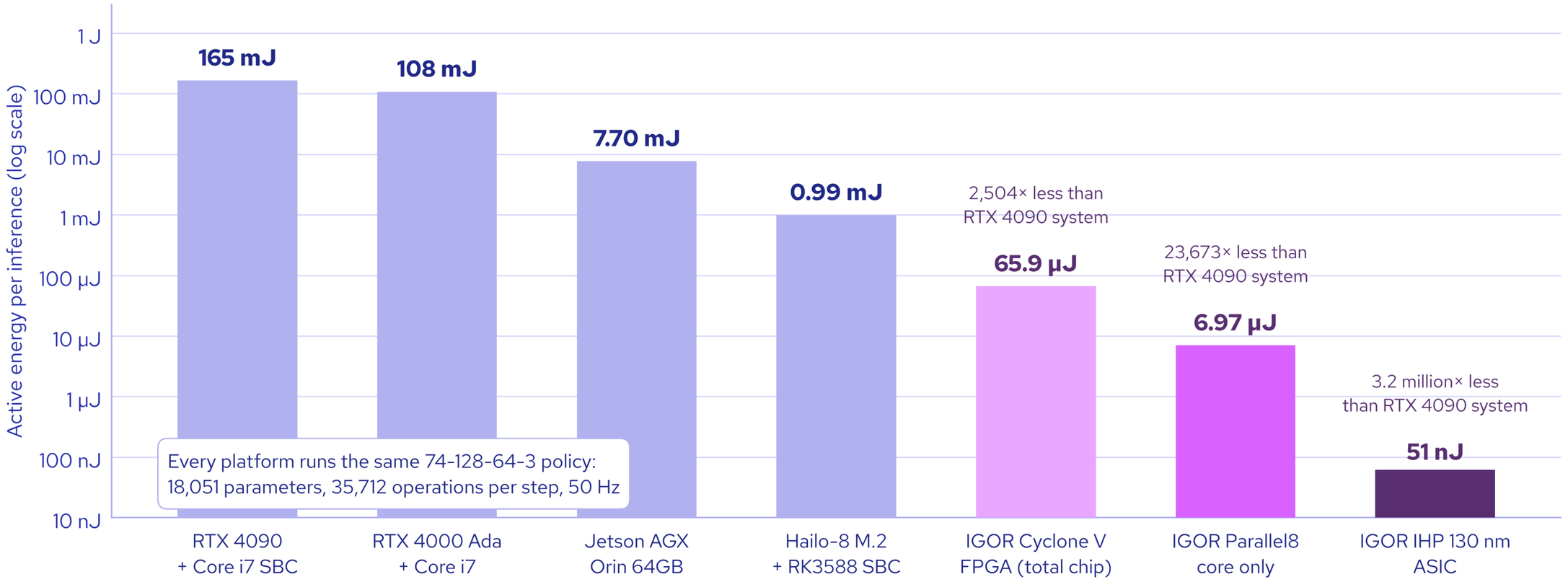
Sources [1] PERURI phase roadmap, PERURI Chip Hackathon 2026 briefing (Pak Erwin) [2] IGOR Testcase and Threat Matrix TC-01 to TC-18, 19 Oct 2026 [3] Investortrust (BPS data via Menperin), 29 Jan 2026, investortrust.id/business/92982 [4] Metro TV News, 1 Aug 2026, metrotvnews.com/read/KdZCAeWJ-peruri-akselerasi-talenta-dan-kedaulatan-teknologi-nasional [5] wafer.space GF180MCU Run 3, crowdsupply.com/wafer-space/gf180mcu-run-3 [6] Regulation (EU) 2023/1230, Annex I Part A, eur-lex.europa.eu/eli/reg/2023/1230/oj [7] AI Omnibus, Reg. (EU) 2026/1744, White & Case, whitecase.com/insight-alert/eu-ai-omnibus-enters-force-amending-ai-act [8] IFR, 30 Sep 2026: 117k sold in 2025, grown at 19% a year, ifr.org/ifr-press-releases/global-sales-of-professional-service-robots-surge-24-percent [9] Interact Analysis, 9 Jan 2026, interactanalysis.com/mobile-robots-market-outpaces-fixed-automation [10] Interact Analysis, Jan 2025, interactanalysis.com/wp-content/uploads/Mobile-Robots-January-2025.pdf

The Edge Wattage Trap: GPUs burn power while idle (ISO 3691-4:2023)



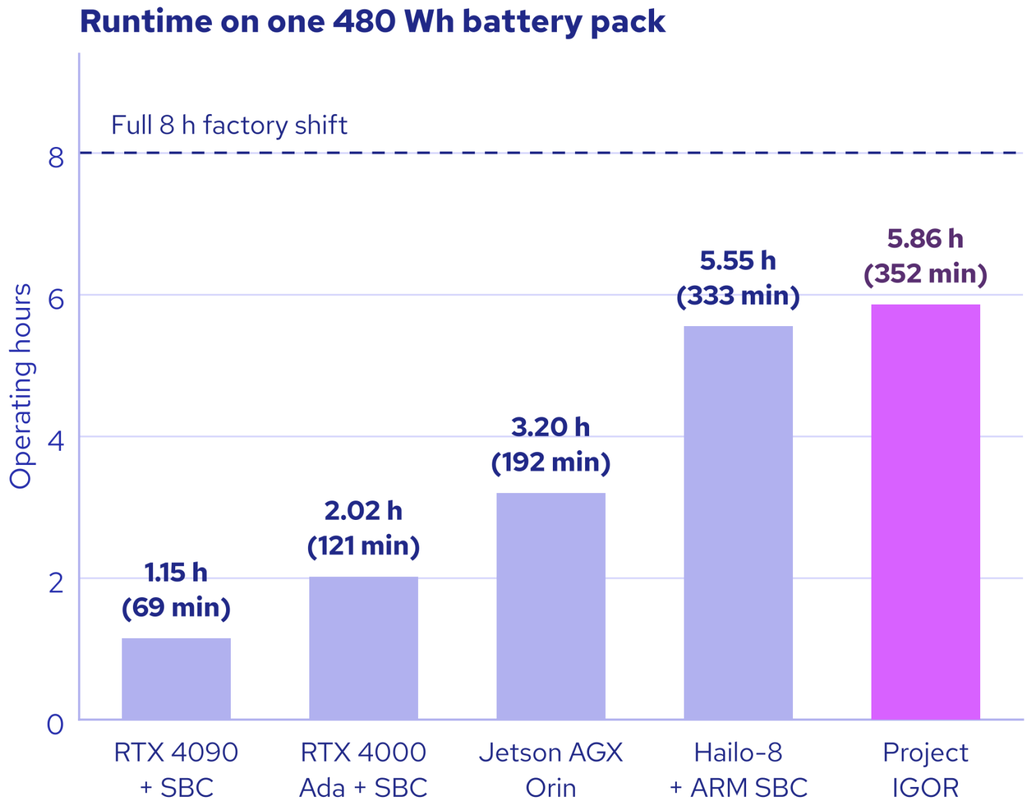
Sources **[1]** IGOR power benchmark report, 14_power_benchmark_nvidia_gpu_sbc_vs_asic_edge_robotics.md, Oct 2026, Table 2: system-level power breakdown (host SBC, DRAM and storage, accelerator, fans, 87% DC-DC efficiency) **[2]** NVIDIA Ada Lovelace Architecture Whitepaper (2023); NVIDIA Jetson AGX Orin Series Data Sheet DS-10654-001 Rev 1.4 (2024) **[3]** Hailo-8 M.2 AI Acceleration Module Datasheet and Architecture Whitepaper (2023) **[4]** Intel Quartus Prime Standard 25.1 Power Analyzer and STA reports, Cyclone V 5CSEBA6U23I7, Parallel8 core: 50.36 mW dynamic, 476.18 mW total, +5.647 ns slack, 6,916 cycles = 138.32 µs (METRICS_AND_POWER.md, BUILD_STATUS.md)

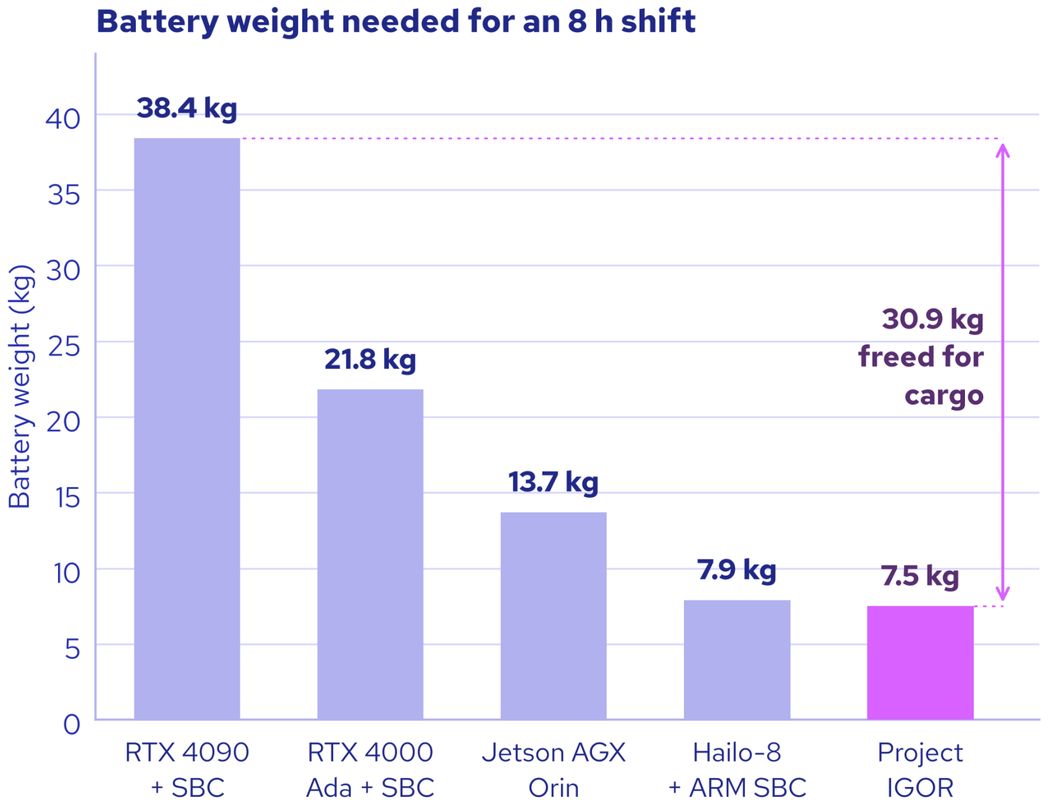
Same policy, same math, far less energy (ISO 3691-4:2023)



Sources **[1]** IGOR power benchmark report, 14_power_benchmark_nvidia_gpu_sbc_vs_asic_edge_robotics.md, Oct 2026, Section 3: apples-to-apples benchmark table (E = active power × latency per step) **[2]** NVIDIA Ada Lovelace Architecture Whitepaper (2023); NVIDIA Jetson AGX Orin Series Data Sheet DS-10654-001 Rev 1.4 (2024) **[3]** Hailo-8 M.2 AI Acceleration Module Datasheet and Architecture Whitepaper (2023) **[4]** Intel Quartus Prime Standard 25.1 Power Analyzer and STA reports, Cyclone V 5CSEBA6U23I7, Parallel8 core: 50.36 mW dynamic, 476.18 mW total, +5.647 ns slack, 6,916 cycles = 138.32 µs (METRICS_AND_POWER.md, BUILD_STATUS.md) **[5]** TinyTapeout IHP 130 nm CMOS5L signoff, tt_um_igorchip: 2.037 mW post-route, 0.916 mm² (docs/ARCHITECTURE.md) **[6]** MLCommons MLPerf Inference Edge v3.1 and v4.0, mlcommons.org/benchmarks/inference-edge

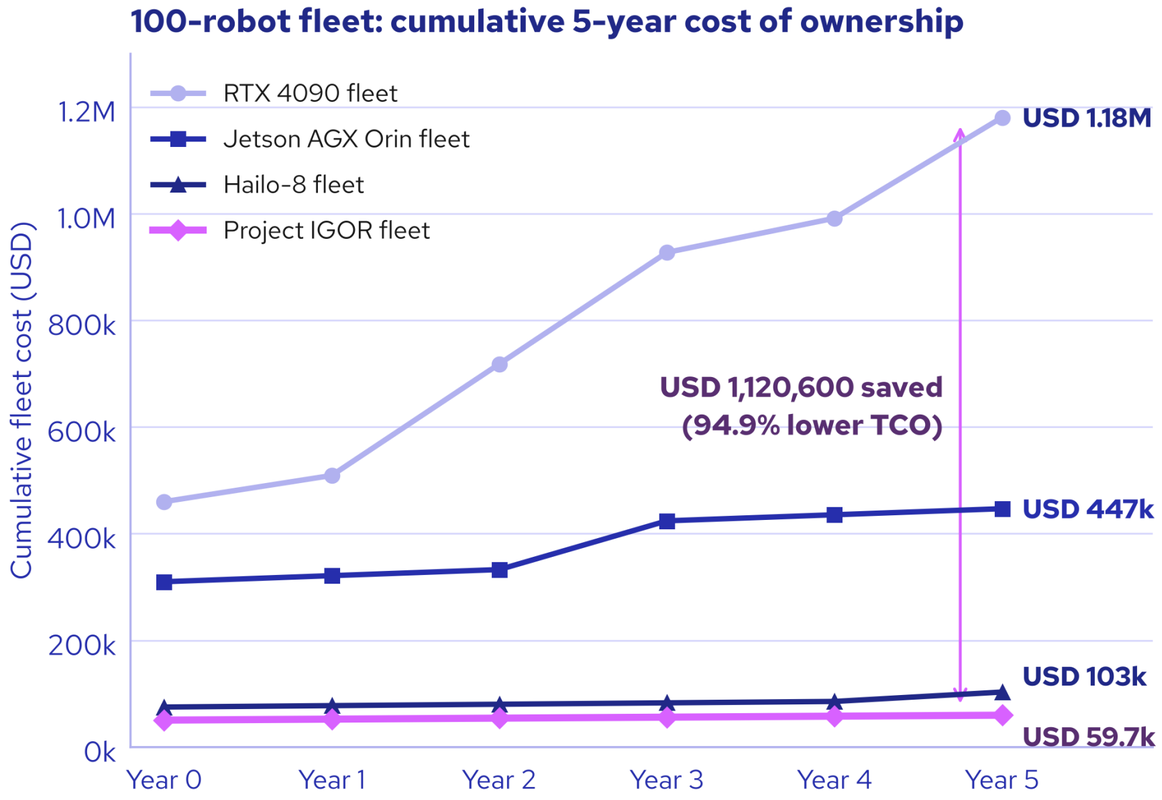
From 69 minutes to nearly 6 hours per charge (ISO 3691-4:2023)

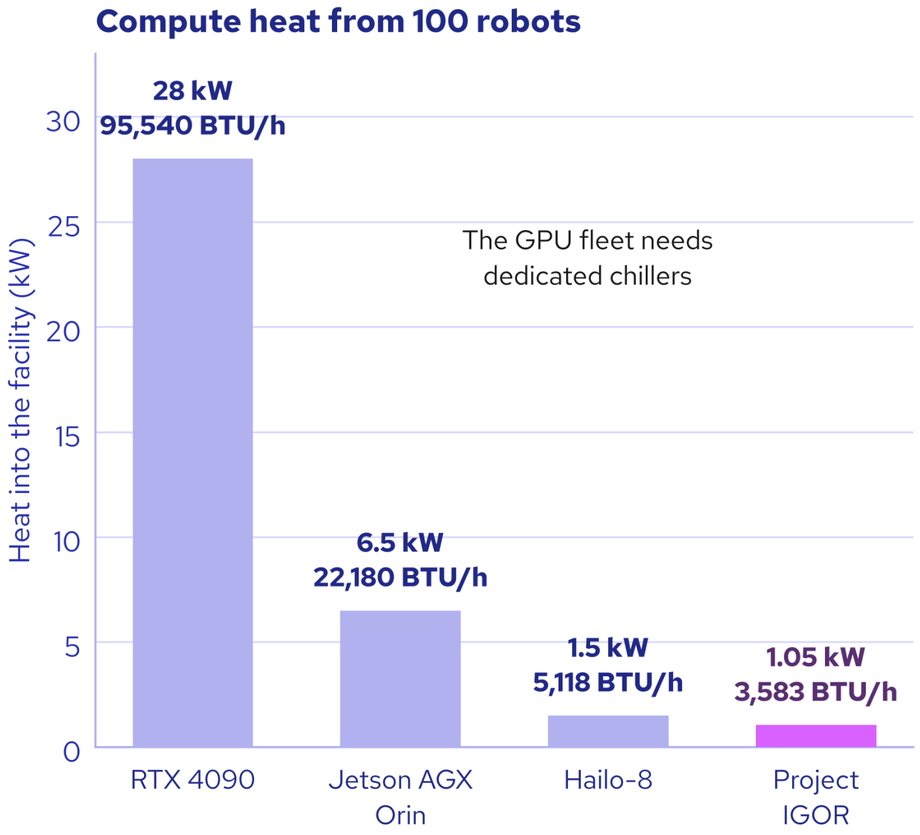




Sources **[1]** IGOR power benchmark report, 14_power_benchmark_nvidia_gpu_sbc_vs_asic_edge_robotics.md, Oct 2026, Table 4: battery life and duty-cycle scaling (24 V 20 Ah LiFePO4, 80% DoD, 55 W motors, sensors and aux baseline) **[2]** Boroumand, A., et al., "Mitigating the Computing Energy Penalty in Autonomous Mobile Robots," IEEE T-RO 39(4), 2891–2908, 2023

100 robots: USD 1.12M saved over 5 years (ISO 3691-4:2023)





Sources **[1]** IGOR power benchmark report, 14_power_benchmark_nvidia_gpu_sbc_vs_asic_edge_robotics.md, Oct 2026, Table 6: 5-year TCO per robot (hardware, battery pack, battery replacements, fan maintenance, electricity and HVAC); yearly curve built from these components **[2]** IGOR power benchmark report, 14_power_benchmark_nvidia_gpu_sbc_vs_asic_edge_robotics.md, Oct 2026, Table 5: fleet scaling, 100 robots 24/7/365, USD 0.15/kWh, chiller COP 3.0

A robot earns back IGOR in under 6 months

Even if only stalls are counted; valued per robot from our Level 4 benchmark and the Karawang minimum wage

Same Level 4 simulated course, same 50 Hz loop [1]



Value per robot per year vs IGOR paid once



4,800 robot-hours a year (2 shifts, 300 days); one robot-hour valued at one Karawang UMK hour, Rp5.89m ÷ 173 h = Rp34k [2]

2.8 months
payback, base case: stalls + speed

5.8 months
if speed adds nothing (stalls only)

11.6 months
at half the stall savings: still under a year

Rp1.27bn a year
for a 50-robot fleet paying Rp295m once

Old FPGA price (Rp15.6m): 7.4 mo base, 15.3 mo stalls only

Not counted: safety, booked at Rp0

0% collisions vs 15–20% for APF [1]; up to Rp62.6m a year per 50-robot fleet at 30% effectiveness [3] [4]

Price headroom

Rp4.5m is 37% of the stall-only first-year value, and payback stays under a year even at half of it

Note Value is from simulation; Rp1.4m integration is an assumption to confirm. ISO 3691-4:2023 requires verified safety functions on AMRs, iso.org/standard/83545.html. **Sources** [1] IGOR ROS 2 and neural navigation README, section 3, Level 4 chaos benchmark (slides 9 and 10); simulation, field pilot to confirm [2] Keputusan Gubernur Jawa Barat No. 561/Kep.862-Kesra/2025: UMK Karawang 2026 Rp5,886,852, Bisnis.com, 28 Dec 2025, ekonomi.bisnis.com/read/20251228/12/1939966 [3] OSHA, Safety Pays, osha.gov/safetypays [4] IGOR expected-loss model: 50 robots, 240k robot-hours, assumed incident rates and costs, 19 Oct 2026

Rp1.1bn buys the evidence before the big spend

A Rp13.7bn programme released in tranches: the first Rp1.1bn (8%) buys paid-pilot evidence before silicon spending opens

2026–27
Secure the IP baseline
Rp0.8bn
Opens
Now: rights, RTL, regression

2027
Paid FPGA pilots
Rp0.3bn
Opens when
Tranche 1 done; a PERURI site first

2028–29
Neural ASIC and first silicon
Rp6.1bn
Opens when
Only after paid pilots

2029–30
Qualification and EU type exam
Rp6.5bn
Opens when
First silicon passes agreed tests

2030 target
Sell ASICs to OEMs
2,000 chips
Opens when
Type exam passed, OEM approval

Rp28.9bn
NPV @12%, base case at the Rp4.5m ASIC price

44.5%
IRR; discounted payback in 2033

Rp13.7bn
peak funding in 2029, Rp17.1bn with a 25% buffer

40%
of base volume is enough to break even

Price lever
Each +Rp1m per chip adds about Rp13bn of NPV, so Rp4.5m balances PERURI's return against customer adoption

Strategic return beyond NPV
A PERURI chip business and a new export line: Rp9.0bn revenue in 2030, Rp49.5bn in 2036

Downside is capped: Rp1.1bn buys the evidence before Rp12.6bn of silicon and qualification spending is committed

Note Price, unit cost, volume, yield and launch dates are assumptions; no FPGA sales enter these returns. Pessimistic (2 years late, 40% of volume, price −20%): NPV −Rp14.3bn; optimistic (1.7× volume): NPV Rp62.3bn. **Sources** **[1]** IGOR ASIC planning model, 22 Oct 2026: Rp4.5m price, Rp0.9m variable cost per chip, 2,000 chips in 2030 rising to 11,000 in 2036, fixed opex Rp2.5bn from launch +10% a year, 10% working capital, no value after 2036 **[2]** UU No. 7/2021 (HPP) Art. 17(1)(b), 22% corporate tax, pajak.go.id/id/artikel/lima-jenis-tarif-pph-badan-yang-wajib-diperhatikan **[3]** Bank Indonesia JISDOR, Rp18,050 on 19 Oct 2026, used as Rp18,000

Fixed 138 µs timing, under 1 W, sized to the job

IGOR today on FPGA vs Jetson Orin NX, Hailo-8 and Infineon AURIX TC367

	IGOR (FPGA, today)	Jetson Orin NX 8GB	Hailo-8	AURIX TC367
Role	**Learned correction + hardware cap**	General AI, vision, planning	CNN vision accelerator	Safety real-time controller
Compute	**17,856 MACs per decision (8 lanes, 50 MHz)**	Up to 70 TOPS sparse INT8 (35 dense)	26 TOPS	2 TriCore cores, 300 MHz
Latency	**138.32 µs fixed (6,916 cycles), no OS; 4.6 ms mean incl. UART round trip**	Linux-scheduled; to measure on the same policy	Not published for this workload	Real-time MCU; not published for this workload
Power	**476 mW typical, 938 mW max (FPGA estimate)**	10–20 W module envelope	2.5 W typical	Under 2 W (device)
Model memory	**35 KiB in on-chip ROM, no write path**	8 GB LPDDR5, writable	Runtime-loaded model	4 MB flash + 672 KB SRAM, ECC
Safety	**E-stop zeroes outputs in 1 clock; 2 ms watchdog; 74 input range checks; RTL clamp ±1.0**	Built by the OEM in software	None for a control role	ASIL-D support, lockstep, self-tests, HSM
Verification	**111,210 RTL oracle vectors; 100% line and branch coverage; not yet certified**	Production silicon	Production silicon	Production silicon
Price	**Rp4.5m (US$250), proposed ASIC**	US$649 list	€195 (M.2, retail)	€23–34 (1–50 pcs)

ASIC silicon proof of concept

The I/O engine chip, not the neural core

0.916 mm²
die on IHP SG13CMOS5L

2.04 mW
typical; 1.62–2.66 mW across corners

PASS
gate-level test of the routed netlist; DRC, LVS, antenna and timing clean

Next on silicon: weight ROM and 9-bit coefficients. 35–64 KiB of weights as SRAM macros alone would be 2.00–3.60 mm², 2.2–3.9× today's die

Fixed by construction
6,916 cycles every time, no OS jitter

20–40× lower power
than a Jetson's 10–20 W, at the FPGA stage

Right-sized
one bounded job, not a TOPS race

Note IGOR power and latency are FPGA figures; the 0.916 mm² and 2.04 mW are for the I/O engine chip, not the neural core, which is not on silicon yet. ASIC power should be lower, to measure. **Sources** **[1]** IGOR deck: control loop, datapath, coverage, clock cycles, FPGA power, speedup and power analysis (slides 8, 13, 17–18, 22–23, 26, 27) **[2]** Mouser and Seeed Studio, Jetson Orin NX 8GB datasheet and list price **[3]** Converge Digest and BuyZero, Hailo-8 specifications and M.2 retail price **[4]** Farnell and Macnica, Infineon AURIX TC367 listings and datasheet

PERURI DIGITAL SUMMIT 2026
HACKATHON

IGOR: Indonesian Gated Onboard Robotics



iSeeDesignITB

AI/Edge Accelerator focus (Topic 3)

PERURI Chip Hackathon 2026

20th October 2026